Anime Clusters and Recommendations

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Contents

1 Introduction **3**
1.1 Background . . . 3
1.2 Problem Description . . . 3
1.3 Dataset Overview . . . 3

2 Methodology **4**
2.1 Data Processing & Challenges . . . 4
2.2 Exploratory Statistical Analysis . . . 4
2.3 Spectral Clustering Analysis . . . 5
2.4 User-Based Collaborative Filtering Recommendation Model . . . 6

3 Evaluation & Results **7**
3.1 Cluster Evaluation . . . 7
3.1.1 Fidelity Metrics Comparison . . . 7
3.1.2 Community Validation Study . . . 8
3.2 Recommendation Evaluation . . . 8
3.3 Cross-Analysis & Key Findings . . . 9
3.4 Cluster Characterization . . . 10
3.5 Future Insight . . . 11

4 Website **11**

Link to the Recommendation System Demonstration Website (Along with Demonstration Video)

1 Introduction

1.1 Background

Anime continues exploding in popularity worldwide, with the industry now worth over $30 billion as of 2025. What started as a cornerstone of Japanese culture has become a global phenomenon, but understanding viewing patterns and preferences remains challenging. Traditional recommendation systems often fail to capture the nuanced relationships between different anime series, missing the complex thematic and stylistic connections that define anime communities. The problem becomes more complex when we consider that anime fans don't just rate shows - they form distinct communities around specific genres, themes, and viewing patterns. A mecha fan's preferences differ significantly from someone who prefers slice-of-life anime, yet current systems often treat all highly-rated shows as equivalent. This creates an opportunity to discover meaningful structure in how users actually interact with anime content.

1.2 Problem Description

We tackle two related but distinct problems in anime analysis. First, can we identify coherent clusters of anime based on user viewing behavior that reveal meaningful thematic or stylistic groupings? Our hypothesis is that spectral clustering applied to user-anime interaction data will uncover community structures that correspond to recognizable anime categories and viewing patterns. Second, we explore whether market-basket analysis can effectively capture the associative patterns in how users watch and enjoy anime. Unlike traditional collaborative filtering, market-basket methods focus on which anime frequently appear together in user viewing histories, potentially revealing "gateway" series and natural progression paths through different anime types. The key insight we're testing is whether these two approaches - clustering for thematic structure and market-basket for behavioral patterns - provide complementary views of the anime ecosystem that can improve both content understanding and recommendation quality.

1.3 Dataset Overview

Our data consists of a comprehensive 15GB collection of anime interaction data obtained from MyAnimeList via Kaggle. The dataset includes detailed information on over 13,000 anime titles, hundreds of thousands of users, and millions of user-anime interactions spanning ratings, favorites, completion status, and detailed reviews. The core components include user-anime interaction records containing ratings (1-10 scale), watch status (completed, watching, dropped, plan to watch), favorite indicators, and timestamp data. We also have anime metadata covering titles, genres, studios, episode counts, and aggregate popularity metrics. Additionally, the dataset contains anime-anime relationship data showing user-curated recommendations and series connections. For computation's sake, we focus our analysis on the 2,000 most popular anime titles - those with sufficient rating history to enable meaningful analysis. This subset represents the most engaged portion of the community while maintaining the diversity needed to discover genuine clustering patterns. A key challenge is the inherent sparsity of the data. Most users rate only a small fraction of available anime, creating a sparse interaction matrix that requires careful handling in both clustering and association rule mining.

2 Methodology

2.1 Data Processing & Challenges

Our 15GB dataset contained substantial missing data, with users rating only small subsets of available anime, creating a sparse interaction matrix with over 95% missing values. We implemented multi-stage filtering requiring users to have rated at least 10 anime and focusing on anime with sufficient rating data. This reduced our working dataset to approximately 1,000,000 active users and 2,000 well-rated anime titles. We normalized all ratings by calculating per-user mean and variance, filtering out users with missing scores. Memory constraints required efficient sparse matrix representations and batch processing techniques optimized for student-level computational resources, particularly critical for spectral clustering eigendecomposition steps. For the recommendation system analysis, computational limitations necessitated using only 20% of the filtered dataset while maintaining representative sampling across users and anime.

2.2 Exploratory Statistical Analysis

We examined whether anime ratings exhibit meaningful structure to support our analytical approach. Related anime (sequels, spinoffs) showed substantially higher rating correlations ($\mu = 0.63$) compared to random anime pairs ($\mu = 0.09$), indicating similar content receives similar ratings. This dramatic difference in correlation distributions validates the presence of content-based rating patterns that extend beyond individual titles to related works.

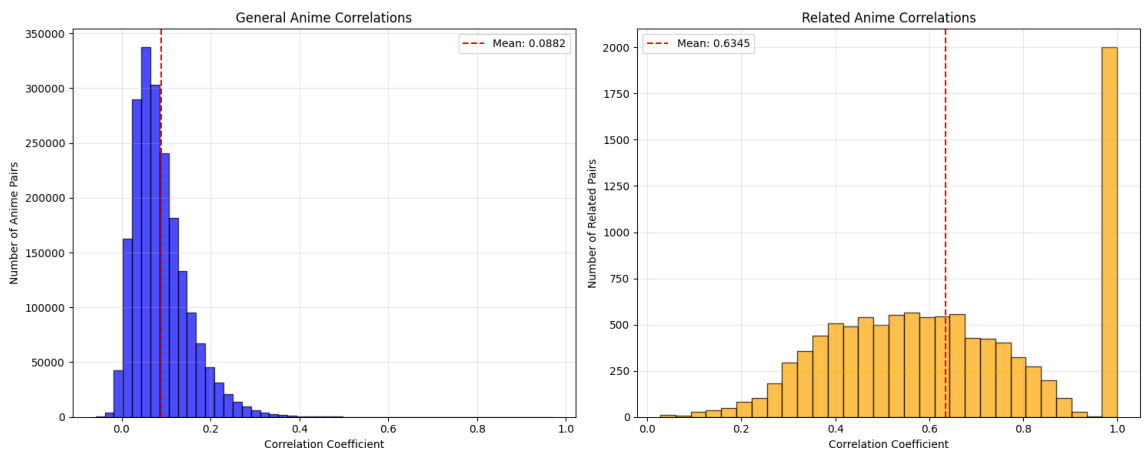

Figure 1: Correlation Between Every Possible Pair of Shows' Scores vs Only Those Related To Eachother

Kolmogorov-Smirnov tests between genre rating distributions revealed significant structural differences, with most genre pairs differing significantly ($p<0.001$). However, some genres like Romance-School ($p=0.947$) and Sci-Fi-Fantasy ($p=0.977$) exhibited similar rating patterns, suggesting overlapping audience preferences. The distribution of scores across genres demonstrates consistent rating behavior within thematic categories, with most genres exhibiting similar central tendencies despite different variance patterns.

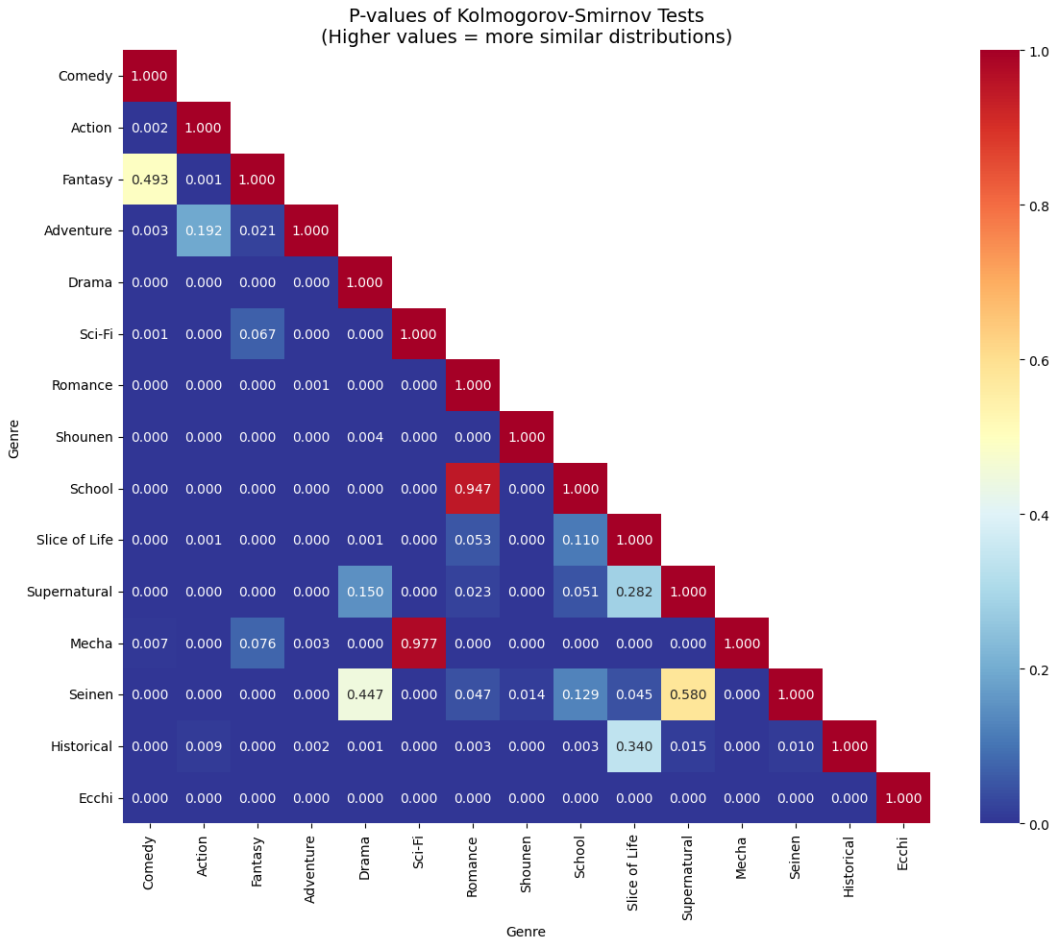

Figure 2: KS Tests Between Each Pair of Anime Genres

These findings validated that clear rating structure exists based on content similarity and thematic relationships, but is complex enough to benefit from algorithmic discovery. The combination of strong content-based correlations and systematic genre-level differences provides the foundation for meaningful clustering approaches that can capture both explicit categorical relationships and implicit user preference patterns.

2.3 Spectral Clustering Analysis

We explored multiple clustering approaches including cover-based clustering using pre-trained ML models on anime images and synopsis-based clustering using text embeddings. Both proved inadequate when evaluated using our fidelity metrics. We developed a comprehensive evaluation framework measuring data purity (genre/studio consistency), relation preservation (keeping related anime grouped), and size balance (entropy calculations preventing lopsided clusters). After iterating over dozens of al-

gorithm and cluster number combinations, we conducted qualitative validation with our team's anime expertise (my anime friends and I) to assess whether groupings aligned with known thematic patterns. Cosine distance-based spectral clustering produced the most interpretable and meaningful results. Our implementation constructs similarity matrices using cosine similarity of user rating vectors with a 0.1 threshold, then performs spectral embedding via normalized graph Laplacian eigendecomposition. K-means clustering in the reduced eigenspace yielded 13 final clusters, validated through genre analysis, rating patterns, and community odd-one-out surveys.

2.4 User-Based Collaborative Filtering Recommendation Model

We implemented a user-based collaborative filtering system using per-item nearest neighbor matching. For each unrated anime title, we identified the training user most similar to the new user (using cosine similarity) among those who rated that specific title, then used that user's rating as our prediction. This per-item approach identified the most relevant user similarity for each anime rather than using fixed global neighbors.

We ranked all unrated anime by predicted ratings and recommended titles with the highest scores. We used a 90/10 train-test split to evaluate recommendation performance. This method addressed the extreme sparsity of anime rating data while capturing user preferences through direct similarity matching rather than averaged collaborative signals.

3 Evaluation & Results

3.1 Cluster Evaluation

3.1.1 Fidelity Metrics Comparison

Our systematic evaluation of clustering methods involved testing multiple algorithms across different cluster numbers, generating extensive quantitative comparisons. We evaluated hierarchical Ward clustering, standard k-means, k-means with PCA preprocessing, and spectral clustering variants across 10-20 clusters each. The results demonstrate clear performance differences between approaches.

Standard k-means methods achieved composite fidelity scores ranging from 0.56-0.58, while hierarchical Ward clustering performed similarly with scores of 0.54-0.56. However, spectral clustering methods outperformed these baselines, with standard spectral clustering achieving consistent performance of approximately 0.60-0.61 across cluster numbers. The rebalanced clustering variants showed intermediate performance around 0.57-0.58, while spectral_robust achieved more modest scores around 0.50-0.54.

Figure 3: Mixed Fidelity Scores Calculated For Each Clustering Method

Among the top-performing methods, manual qualitative analysis was conducted to assess interpretability and thematic coherence. The standard spectral configuration with 13 clusters was selected based on both its superior fidelity score (0.61) and manual validation with our team's anime expertise. This configuration achieved optimal balance across our evaluation metrics: strong size distribution, reasonable preservation of anime relationships, and coherent thematic groupings that aligned with recognizable anime categories. The 13-cluster solution provided sufficient granularity to capture distinct anime communities while maintaining interpretable and actionable cluster boundaries.

3.1.2 Community Validation Study

To validate whether our computational clusters align with human perception of anime similarity, we implemented an odd-one-out survey distributed to anime fan communities. The study design uses k-medoids selection to identify representative anime from each cluster, creating 26 questions where participants identify the mismatched anime among five options. We created three versions of the survey to ensure comprehensive coverage of all anime titles without repetition, with each cluster represented in two questions per version.

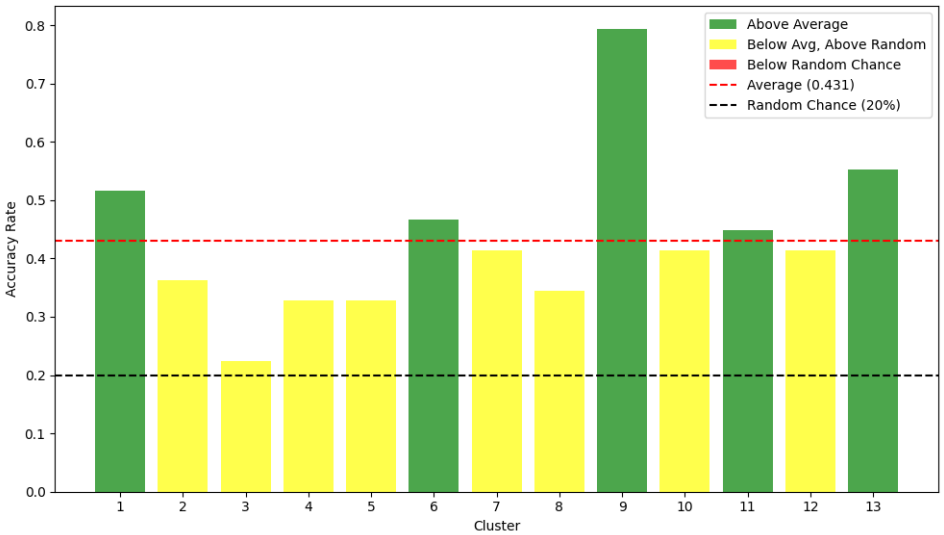

Figure 4: Odd-One-Out Accuracy per Cluster

Results from 29 responses demonstrated validation of our clustering approach. Participants achieved an overall accuracy of 43.1%, significantly above the 20% random chance baseline. Statistical significance testing using binomial tests confirmed that results were highly significant ($p < 0.000001$) with a large effect size (Cohen's $h = 0.505$). All survey versions showed statistically significant performance, with accuracy ranging from 33.8% to 50.0%. Question-level analysis revealed variation from 10.3% to 86.2% accuracy across individual questions. Overall, 92.3% of questions (24 out of 26) performed above random chance, demonstrating that our spectral clustering captured meaningful thematic relationships that aligned with human perception of anime similarity.

3.2 Recommendation Evaluation

Our recommendation evaluation uses a rigorous train-validation-test methodology focusing on active users with substantial viewing history. We filtered for users with at least 50 completed anime, ensuring sufficient data for model development and evaluation. The dataset was split into 18% training, 2% validation, and 1% testing sets, providing robust separation between model development and performance assessment.

For evaluation, we employed cosine similarity comparisons between our collaborative filtering model's predictions and actual user preferences. The recommendation system generates predicted

preference vectors for test users, which are then compared against their actual rating patterns using cosine similarity metrics, where higher values indicate better alignment with user preferences.

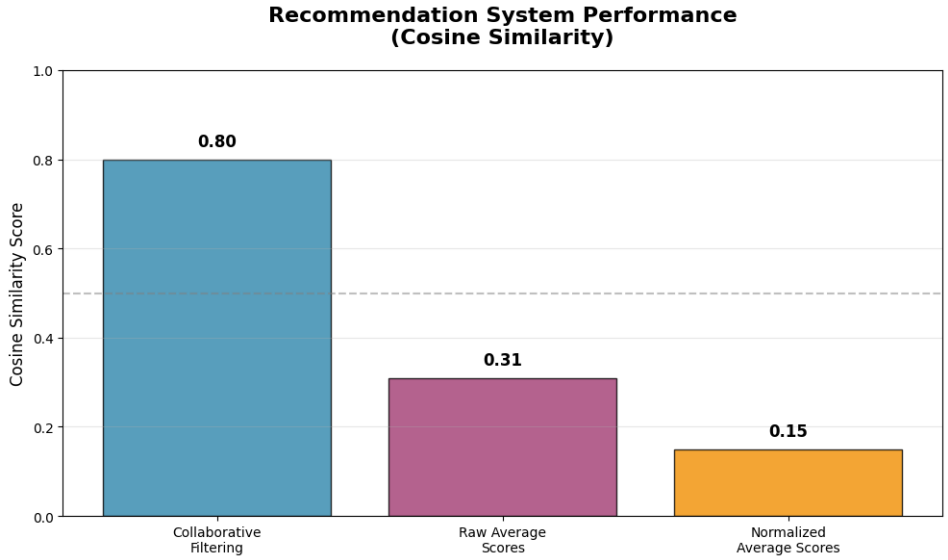

Figure 5: Cosine Similarity Scores for Our Model vs Baseline Predictions

Our baseline comparisons include two fundamental approaches: raw average scores for each anime title, and normalized average scores that account for rating variance across titles. These baselines represent simple popularity-based recommendation strategies against which we measure the effectiveness of our user-based collaborative filtering approach. This evaluation framework directly measures our system's ability to capture personalized user preferences beyond basic popularity trends, providing a clear assessment of the collaborative filtering model's performance in predicting individual viewing preferences.

3.3 Cross-Analysis & Key Findings

To evaluate whether our recommendation system captured the cluster structure discovered through spectral clustering, we conducted a cross-analysis experiment. We simulated user preferences by assigning high ratings to cluster representatives from individual clusters while giving low ratings to representatives from all other clusters, then analyzed whether the resulting recommendations aligned with the expected cluster boundaries.

The recommendation system generated 130 recommendations across all 13 clusters, with 117 existing in our original dataset for validation. When evaluated against ground-truth cluster assignments, our system achieved an overall precision of 19.2%, representing a 2.5x improvement over the 7.7% random baseline. Performance varied significantly across clusters, with precision ranging from 0% to 50%. Clusters 2 (atmospheric fantasy-adventure) and 11 (supernatural mystery) demonstrated the strongest alignment at 50% precision each, while clusters 3, 7, 12, and 13 showed no meaningful cluster awareness (0% precision).

The cross-analysis revealed fundamental insights about the relationship between thematic clustering and collaborative filtering behavior. Certain clusters, particularly those with coherent thematic identity like Romance/Drama (Cluster 1) and Prestige content (Cluster 2), translate more effectively into recommendation patterns. Conversely, clusters with more complex or nuanced thematic boundaries showed poor collaborative filtering alignment, suggesting these groupings reflect subtle content relationships not captured by typical user rating behaviors. This validates our hypothesis that meaningful anime structure exists beyond simple genre categorization, while revealing that collaborative filtering systems naturally capture only a subset of these thematic relationships.

3.4 Cluster Characterization

Our 13 spectral clusters demonstrate meaningful thematic and structural patterns in the anime landscape. Each cluster exhibits distinct characteristics in terms of genre composition, production type distribution (TV series, OVAs, movies), and average rating patterns.

Cluster	Refined Theme	Top Genres	Key Characteristics
1	School Romances	Romance, Comedy, School	High school romance/harem with comedy
2	Prestige Movies/Shows	Drama, Action, Fantasy	Critically acclaimed, high production
3	Action-Romance Mix	Comedy, Action, Fantasy	Mixed action, fantasy, romantic elements
4	Classic/Nostalgic Anime	Comedy, Action, Romance	Well-known series from earlier eras
5	Modern Isekai/Fantasy	Fantasy, Comedy, Action	Recent isekai/fantasy with comedy
6	Top Popular Titles	Action, Shounen, Drama	Blockbuster anime with mass appeal
7	Women's Preferences	Romance, Drama	BL, shoujo, otome content
8	Character-Driven Series	Comedy, Shounen, Action	Long-running with character development
9	Kids/Family Movies	Adventure, Fantasy, Action	Family-friendly adventure content
10	Sequels & Spinoffs	Comedy, Action, Shounen	Continuation content and extensions
11	Side Content (OVAs/Specials)	Comedy, Action, Supernatural	OVAs, specials, supplementary episodes
12	Dark/Mature Anime	Action, Fantasy, Mystery	Darker, Mature, Psychological content
13	Slice of Life (CGDCT)	Comedy, SoL, School	"Cute Girls Doing Cute Things"

Table 1: Breakdown of each Cluster by Defining Characteristics

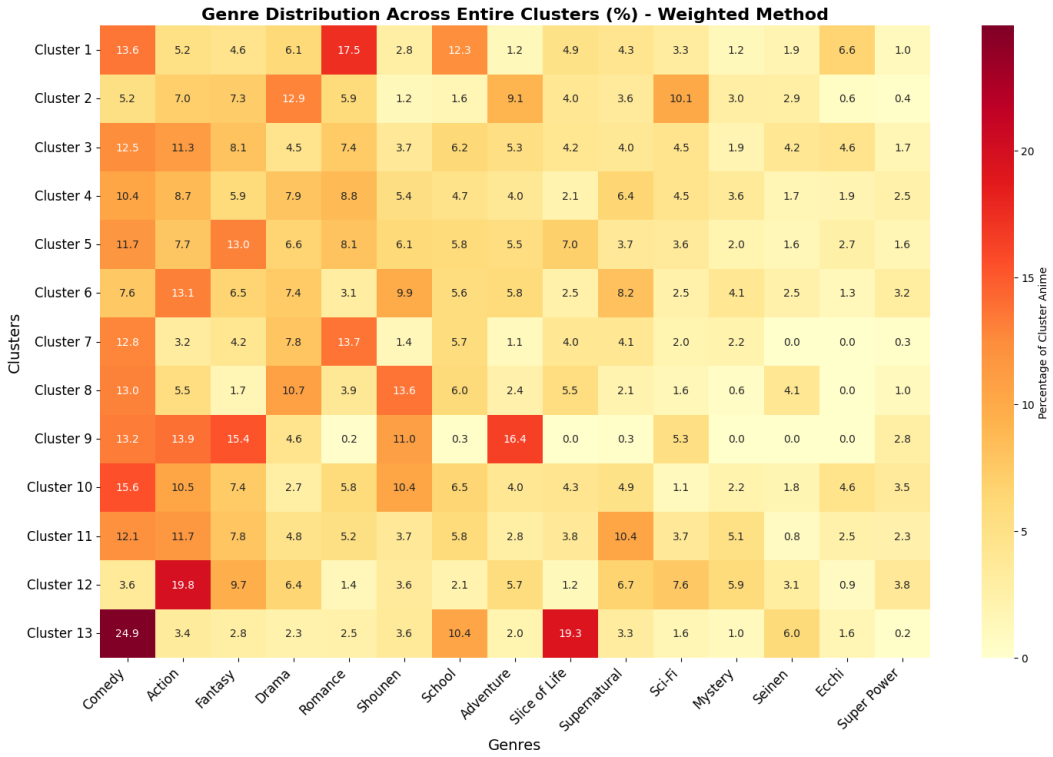

Figure 6: Genre Breakdown Per Cluster

3.5 Future Insight

These findings suggest several directions for future investigation, including temporal analysis of community evolution and multimodal integration of textual features to enhance both clustering fidelity and recommendation performance through hybrid similarity metrics.

4 Website

Attached at the table of contents is a link to a basic website for our recommendation model, along with a demonstration video under the "demonstration folder". The web application provides an interactive interface for users to experience our collaborative filtering recommendation system firsthand.

The website presents users with an extensive list of anime titles where they can input their personal ratings on a 1-10 scale for shows they have watched, leaving blank entries for unfamiliar content. The interface accommodates diverse viewing histories by including popular mainstream titles alongside niche series, allowing users to accurately represent their preferences across different anime categories and genres.

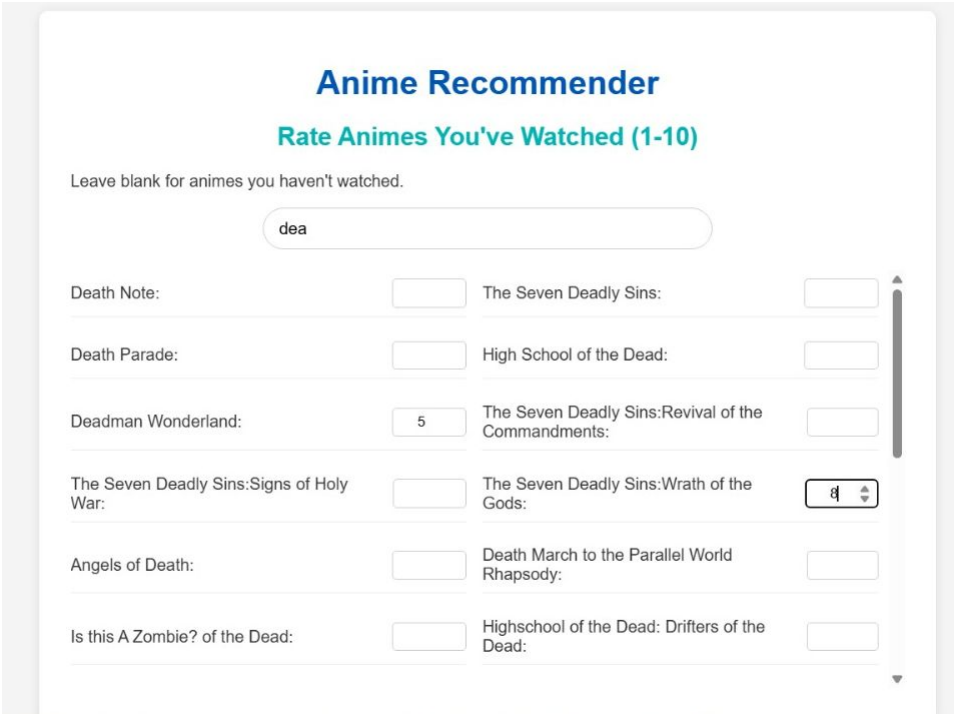

Figure 7: User Score Inputting Interface

Upon rating their watched anime, users specify the desired number of recommendations (up to 50) and receive personalized suggestions generated by our trained collaborative filtering model. The system processes the user's rating profile in real-time, identifying similar users within our dataset and generating recommendations based on the viewing patterns and preferences of their nearest neighbors. The output presents a ranked list of recommended anime titles, demonstrating the practical application of our research findings in a user-friendly format that could be deployed in real-world recommendation scenarios.

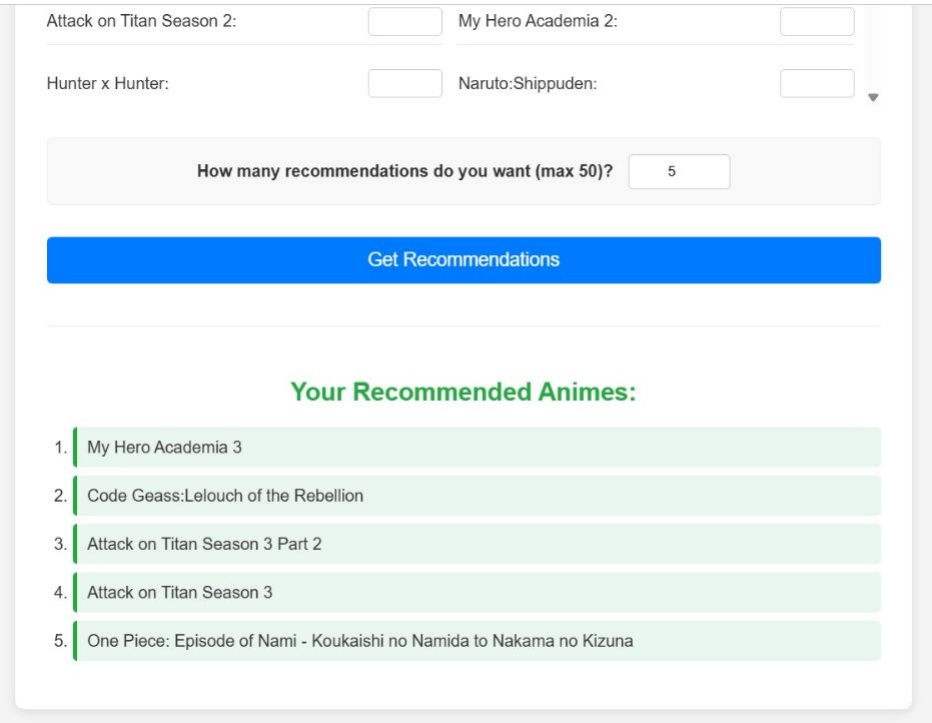

Figure 8: Recommendation Interface